

## Code-Layout, Readability and Reusability

|                      |                                                |
|----------------------|------------------------------------------------|
| <b>Date</b>          | 07 July 2026                                   |
| <b>Team ID</b>       |                                                |
| <b>Project Name</b>  | HDI Predictor using Machine Learning and Flask |
| <b>Maximum Marks</b> | 5 Marks                                        |

### Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

### Code Layout Checklist:

| S.No | Code Quality Parameter    | Description                                   | Followed (Yes / No / Partial) | Remarks                                  |
|------|---------------------------|-----------------------------------------------|-------------------------------|------------------------------------------|
| 1    | Consistent Indentation    | Uniform spacing/tabs used throughout          | Yes                           | Well-structured indentation              |
| 2    | Proper File Structure     | Files and folders are logically organized     | Yes                           | Logically organized files and folders    |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly       | Yes                           | Variables are descriptive and purposeful |
| 4    | Function / Method Names   | Functions are descriptively named             | Yes                           | Functions are named descriptively        |
| 5    | Code Comments             | Inline and block comments explain logic       | Yes                           | Includes inline and block comments       |
| 6    | Modular Design            | Code is split into reusable functions/modules | Yes                           | Code is split into reusable modules      |
| 7    | No Redundant Code         | Duplicate or unused code is removed           | Yes                           | No duplicate or unused code              |
| 8    | Error Handling            | Exceptions and errors are handled gracefully  | Yes                           | Exceptions handled gracefully            |

### Reusable Components / Modules:

| S.No | Component / Module Name | Language / Technology | Where Reused | Reusability Level (High / Medium / Low) |
|------|-------------------------|-----------------------|--------------|-----------------------------------------|
|------|-------------------------|-----------------------|--------------|-----------------------------------------|

|   |                   |                    |               |      |
|---|-------------------|--------------------|---------------|------|
| 1 | Prediction Engine | Flask/Scikit-Learn | Backend logic | High |
|---|-------------------|--------------------|---------------|------|

|   |                     |                     |                       |        |
|---|---------------------|---------------------|-----------------------|--------|
| 2 | User Interface Form | HTML/Flask/Jinja2   | Frontend inputs       | High   |
| 3 | CSV Processor       | Python/Pandas       | Batch processing      | Medium |
| 4 | Data Visualizer     | JavaScript/Chart.js | Results visualization | Medium |
| 5 | Auto-fill Service   | Python/CSV          | Data retrieval        | Medium |

Overall Code Quality Assessment:

| Aspect                   | Rating (1-5) | Comments                         |
|--------------------------|--------------|----------------------------------|
| Code Layout & Structure  | 5            | Excellent organization           |
| Readability              | 5            | Code is clean and well-commented |
| Reusability              | 4            | Modular design supports reuse    |
| Documentation / Comments | 5            | Well-documented code             |
| Overall Score            | 5            | Excellent project implementation |